

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
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CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week7 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number:14.3(Present assignment number)/24(Total number of assignments)			
Q.No.	Question	Expected Time to complete	
1	Lab 14: Web Design Applications Using AI Assistance HTML, CSS, and JavaScript Generation with GitHub Copilot Lab Objectives The objectives of this laboratory exercise are to:	Week7 – Wednesday	

	• Design functional and visually appealing web applications using HTML, CSS, and JavaScript. • Use AI-assisted tools to accelerate front-end development. • Apply responsive, accessible, and interactive web design principles. • Customize and validate AI-generated UI code for real-world applications.	
	Learning Outcomes After completing this lab, students will be able to: • Generate structured HTML layouts using AI assistance. • Apply responsive CSS techniques such as Flexbox and Grid. • Implement interactive client-side features using JavaScript. • Critically review and improve AI-generated front-end code.	
	Tools Required • Visual Studio Code (VS Code) • GitHub Copilot Extension • Web Browser (Chrome / Firefox) • HTML5, CSS3, JavaScript (ES6)	
	Task 1: AI-Assisted Personal Portfolio Website Scenario A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization. Tasks 1. Use GitHub Copilot to generate a semantic HTML structure with sections: ○ Home ○ About ○ Projects ○ Contact 2. Ask Copilot to suggest responsive CSS using Flexbox or Grid. 3. Customize AI-generated CSS to add: ○ Hover effects on project cards ○ Smooth scrolling navigation Expected Outcome • Clean and well-structured portfolio website • Responsive layout for desktop and mobile screens • Interactive project cards	
	Task 2: AI-Generated Restaurant Landing Page Scenario A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components. Tasks 1. Use Copilot to create a navigation bar with links: ○ Home ○ Menu ○ Gallery ○ Contact	

	2. Generate a menu section styled with CSS cards. 3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions. 4. Customize animations or transitions for better user experience. Expected Outcome • Fully functional restaurant landing page • Responsive navigation bar • Working image slider with smooth transitions	
	Task 3: AI-Powered Event Registration Form Scenario SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input. Tasks 1. Ask Copilot to generate an HTML form with the following fields: ○ Name ○ Email ○ Phone Number ○ Department ○ Event Selection 2. Use AI assistance to style the form using CSS for readability and accessibility. 3. Implement JavaScript validation using Copilot: ○ Email format validation ○ Phone number length validation ○ Required field checks Expected Outcome • User-friendly and visually appealing form • Real-time validation messages • Prevention of invalid submissions	
	Task 4: AI-Assisted E-Commerce Product Page Scenario A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality. Tasks 1. Use Copilot to generate a grid-based product catalog using HTML and CSS. 2. Implement an “Add to Cart” feature using JavaScript. 3. Modify AI-generated code to include: ○ Cart item counter at the top-right corner ○ Visual feedback when a product is added Expected Outcome • Functional product catalog • Working cart counter • Interactive user experience	
	Task 5: Create a Responsive Web Page Layout Instructions: • Design a basic web page layout with a header, main content area, and footer using HTML and CSS.	

	• Use AI to assist in generating responsive CSS for different screen sizes. • Ensure the layout is clean and visually organized. Expected Output: • A responsive web page with: • Header with navigation links • Main section with placeholder text/images • Footer with copyright or contact info • Layout adapts correctly to desktop, tablet, and mobile screen sizes. Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.	
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